

MISSOULA, MT, USA

WMO: 727730

Lat: 46.921N

Lon: 114.093W

Elev: 3192

StdP: 13.08

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 24153

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-1.7	4.3	-10.3	3.6	-0.7	-3.3	5.2	5.4	23.9	22.2	20.5	23.7	3.8	120	0.599

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.7	93.3	61.6	90.0	61.1	86.5	60.2	64.7	85.2	63.1	83.9	61.6	82.1	8.6	10

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
58.2	81.6	68.5	56.1	75.4	67.8	53.9	69.6	66.6	31.2	85.2	30.0	83.4	28.8	82.0	71.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.8	18.3	16.1	DB	-10.1	98.9	8.0	2.7	-15.9	100.8	-20.6	102.4	-25.0	104.0	-30.8	105.9	
			WB	-10.6	67.8	7.9	1.8	-16.3	69.1	-20.9	70.1	-25.3	71.1	-31.0	72.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.9	25.7	29.1	37.7	44.7	53.3	60.4	69.7	67.9	58.2	44.6	33.1	25.1	(6)
	DBStd	17.16	9.70	9.28	7.39	6.36	7.02	7.01	5.89	5.56	7.28	7.26	8.25	9.35	(7)
	HDD50	3450	755	586	382	184	46	6	0	0	16	195	507	773	(8)
	HDD65	7331	1220	1006	845	610	367	172	25	37	224	631	956	1238	(9)
	CDD50	1947	0	0	1	24	147	318	612	554	261	29	1	0	(10)
	CDD65	352	0	0	0	0	3	33	172	125	19	0	0	0	(11)
	CDH74	6109	0	0	0	14	175	654	2707	2059	491	9	0	0	(12)
	CDH80	2633	0	0	0	1	36	220	1306	923	146	1	0	0	(13)
	Wind	WSAvg	5.0	3.5	4.5	5.5	6.3	6.1	5.8	5.3	4.6	4.1	3.9	3.8	(14)
	Precipitation	PrecAvg	13.6	1.1	0.8	0.9	1.1	1.7	1.9	0.9	1.0	0.9	1.0	1.1	(15)
(16) Precipitation	PrecMax	21.8	3.0	2.5	2.1	3.1	7.4	4.2	3.1	3.3	3.6	3.5	3.0	4.4	(16)
	PrecMin	9.0	0.2	0.2	0.2	0.1	0.2	0.4	0.0	0.0	0.0	0.0	0.2	0.2	(17)
	PrecStd	2.7	0.6	0.6	0.4	0.6	1.1	0.9	0.7	0.8	0.8	0.7	0.7	0.7	(18)
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.4	52.0	65.1	75.8	84.4	91.7	98.8	95.8	89.5	74.4	57.6	46.8	(19)
		MCWB	38.9	42.1	47.3	53.3	58.4	62.8	64.2	62.1	60.3	54.7	49.0	40.6	(20)
	2%	DB	42.3	46.3	58.2	69.3	78.9	86.2	94.6	92.3	84.2	67.1	51.7	42.1	(21)
		MCWB	37.3	39.3	44.7	50.7	56.5	60.9	62.6	60.9	58.4	51.5	44.0	37.6	(22)
	5%	DB	39.4	42.8	53.6	63.8	74.1	81.3	91.2	89.2	79.9	61.4	47.9	38.9	(23)
		MCWB	35.5	37.1	42.6	47.9	54.8	58.9	61.7	60.4	57.0	48.8	41.9	35.3	(24)
	10%	DB	36.8	39.6	49.7	58.1	68.9	76.9	87.7	85.7	75.2	56.7	44.3	36.1	(25)
		MCWB	33.5	34.9	40.5	45.3	52.5	57.3	60.8	59.5	55.5	46.9	39.6	33.2	(26)
	0.4%	WB	40.4	43.5	48.8	55.1	60.4	65.2	68.0	65.6	63.2	56.2	49.9	41.6	(27)
		MCDB	44.3	49.9	61.1	73.1	78.3	84.5	89.9	86.8	79.0	70.5	56.5	45.6	(28)
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	37.8	40.3	46.0	51.4	58.1	62.5	65.6	63.4	60.5	52.9	44.9	38.1	(29)
		MCDB	41.3	45.2	56.2	66.5	75.2	82.0	86.5	85.4	79.1	64.3	50.4	41.3	(30)
	5%	WB	35.6	37.6	43.6	48.9	55.9	60.4	63.9	62.0	58.6	50.3	42.4	35.7	(31)
		MCDB	38.9	41.6	52.6	61.5	71.2	77.7	84.9	83.4	76.4	59.4	47.1	38.2	(32)
	10%	WB	34.0	35.5	41.3	46.5	53.7	58.7	62.3	60.5	56.4	47.8	40.3	33.5	(33)
		MCDB	36.3	38.7	48.8	56.9	66.9	73.7	83.1	81.0	73.0	55.6	44.0	35.5	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	12.3	14.8	19.6	22.4	24.7	25.6	32.7	32.7	28.5	21.5	14.7	11.4	(35)
		MCDBR	14.5	17.8	27.4	34.1	35.2	35.0	38.4	39.0	37.4	30.5	20.1	13.1	(36)
		MCWBR	10.2	11.8	15.7	17.7	16.0	14.3	13.5	14.3	16.0	16.6	13.5	9.9	(37)
	5% WB	MCDBR	13.9	16.5	25.4	30.9	31.1	31.3	33.4	34.4	33.9	26.9	18.5	12.1	(38)
		MCWBR	10.2	11.3	14.9	16.5	14.5	13.5	12.8	13.1	14.8	14.9	12.8	9.5	(39)
(40) Clear-Sky Solar Irradiance	taub		0.261	0.273	0.296	0.329	0.339	0.348	0.351	0.384	0.343	0.311	0.284	0.264	(40)
	taud		2.425	2.398	2.444	2.412	2.449	2.454	2.438	2.339	2.434	2.526	2.527	2.458	(41)
	Ebn at Noon		269	288	297	294	293	290	287	273	276	271	259	255	(42)
	Edh at Noon		21	27	30	34	34	35	35	36	30	24	19	18	(43)
(44) All-Sky Solar Radiation	RadAvg		398	666	1033	1455	1747	1992	2311	1926	1378	826	437	311	(44)
	RadStd		44	69	95	116	141	168	143	113	139	83	43	31	(45)

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	-0.88	N/A	N/A	+99	N/A	(46)
(47)	Regional (7 neighbors)	N/A	N/A	N/A	N/A	-0.63	-1.03	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page